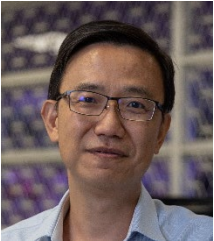
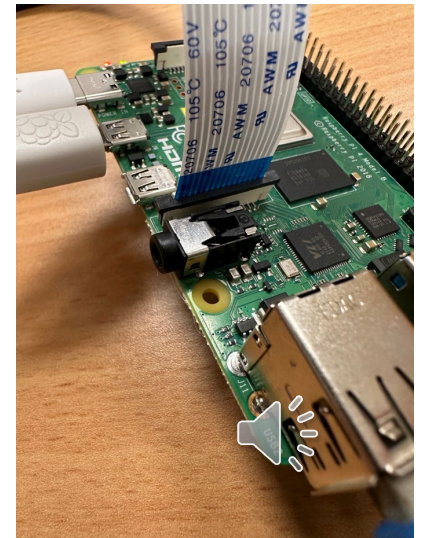


SC2079 MULTIDISCIPLINARY DESIGN PROJECT

RPI and Image Recognition



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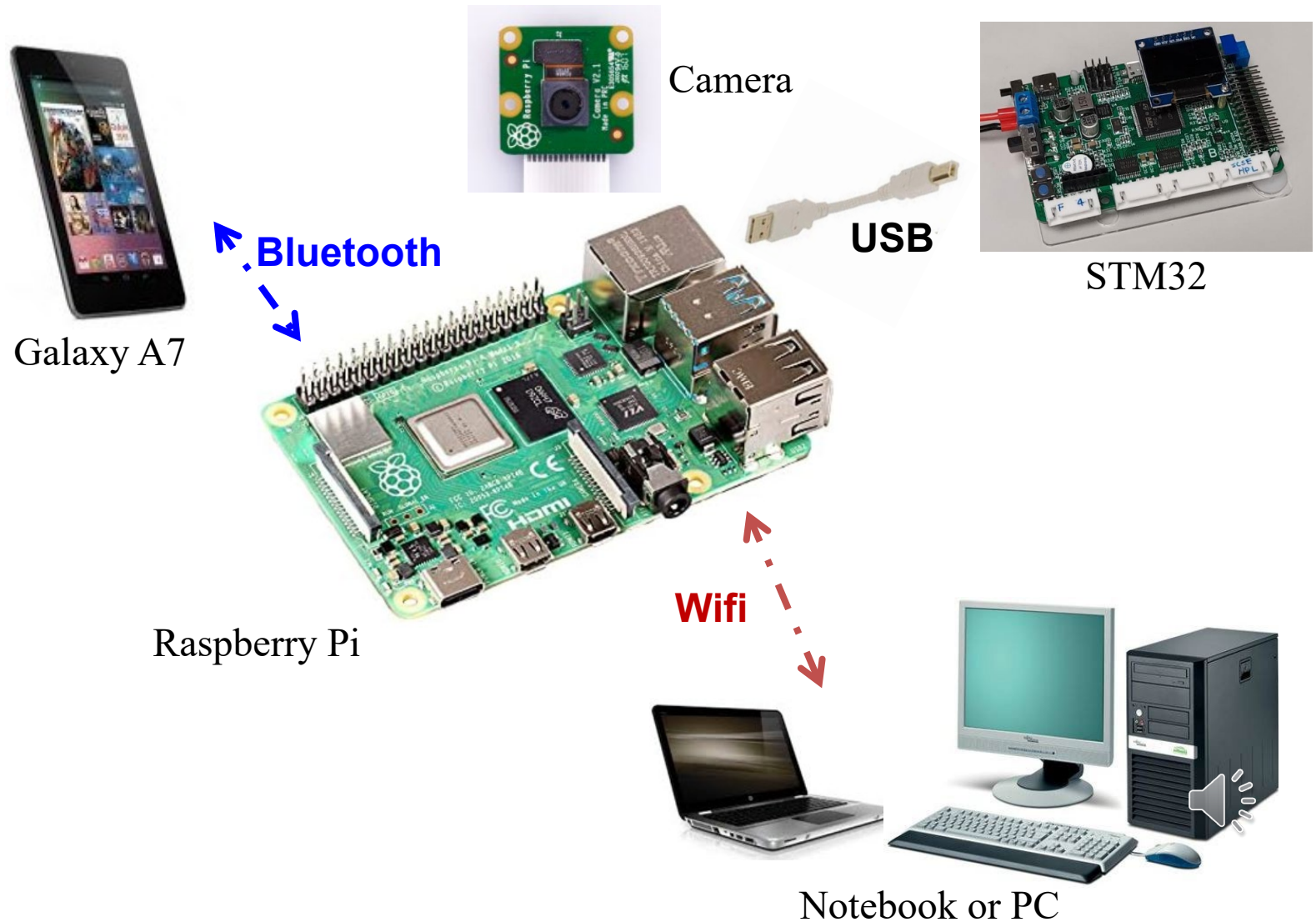


Functions of RPi

- The RPi is the command centre of the MDP robotic system. Main functions.
 - Coordinates and pass information between various modules such as the Android Tablet, STM32 Micro-controller board and the PC.
 - May also performs image related functions such as image capture, processing and recognition.
- Typical system communication setup are as follows.
 - RPi to PC => WIFI
 - RPi to STM32 => Virtual COM Port (ACM) over USB
 - RPi to Android Tablet => Classic BT (SPP).

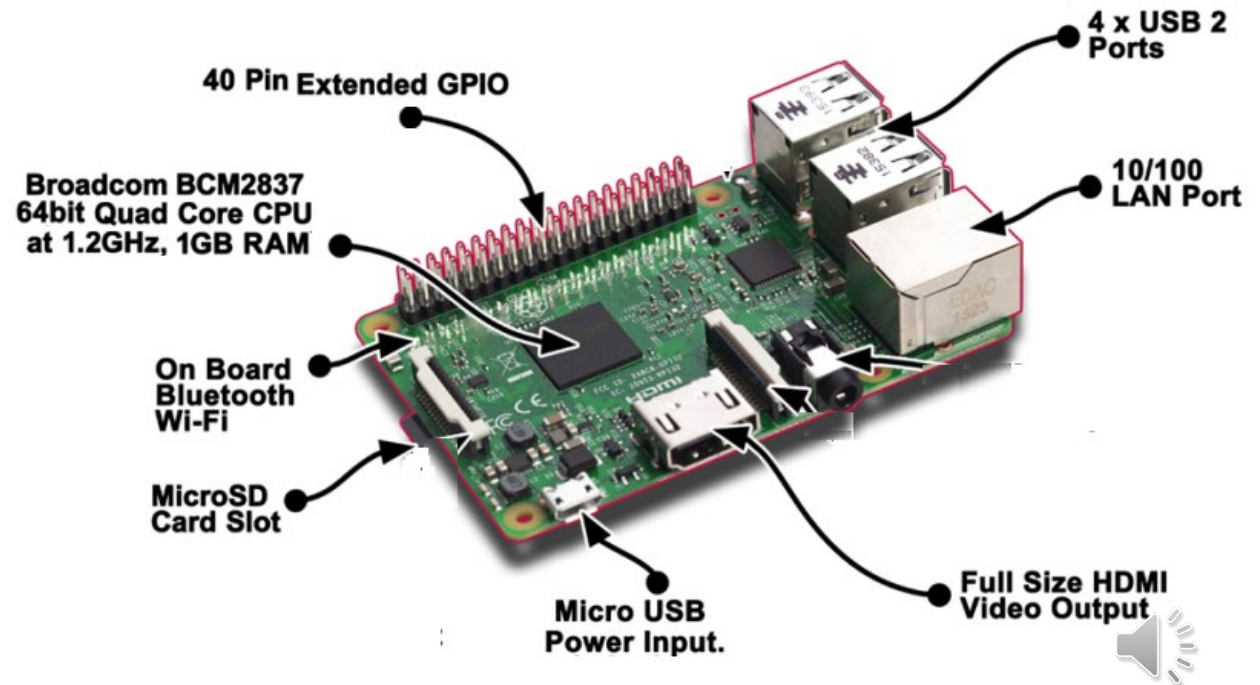


System Interfaces



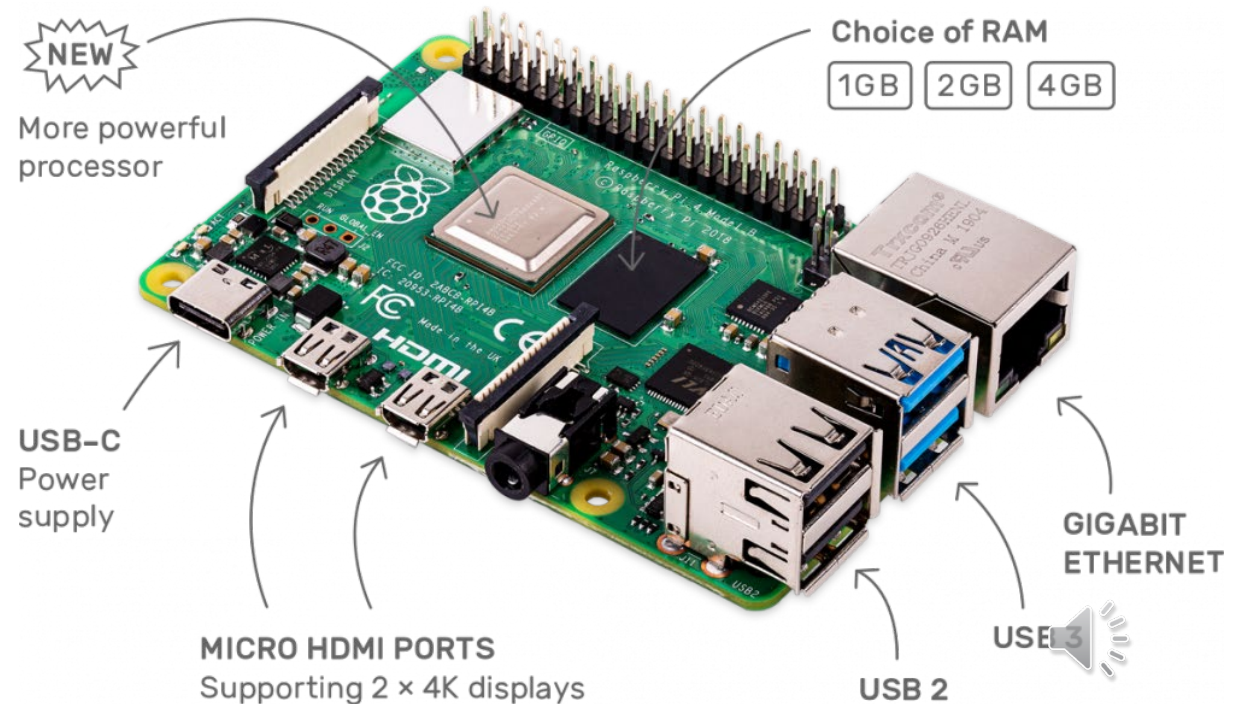
RPI3

- Based on ARM processor (64-bit quad-core Cortex A53 CPU)
- On board Bluetooth and Wi-Fi



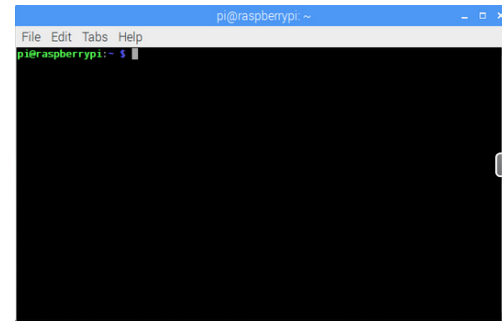
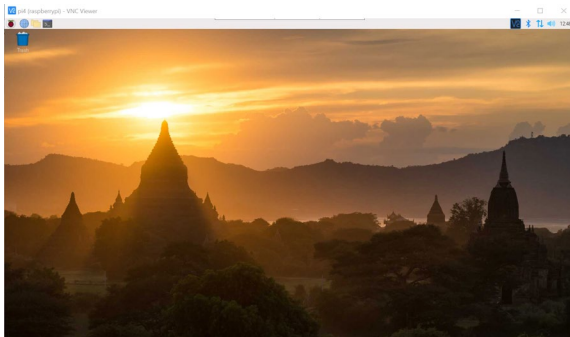
RPI4

- Based on ARM processor (64-bit quad-core Cortex A72 CPU)
- On board Bluetooth and Wi-Fi
- Larger SDRAM



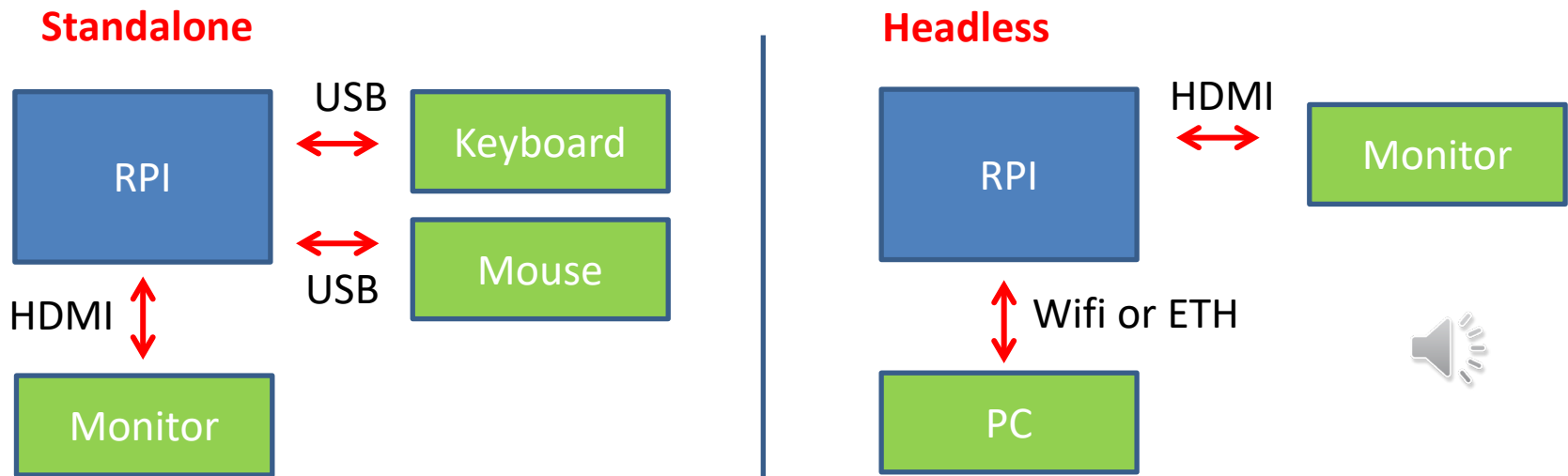
Raspberry Pi OS

- Used to be call Raspbian OS, renamed to Raspberry Pi OS. Based on Linux (Debian).
- Has both command line-based interface or GUI based desktop.
- Remote access can be done via **SSH** (command line) or **VNC** (GUI).
- Text based interface is typically more efficient and allows automation via script files. But requires knowledge of basic linux commands and editor (vi, nano). This is the preferred method.
- GUI based interface is more intuitive so useful during initial phase when user is not familiar with Linux.
- Latest version is v12 (Bookworm) but **Buster (v10)** is preferred due to better compatibility with sw and hw.



Development System Setup

- RPI can be setup as a standalone computer with keyboard, mouse and monitor.
- This will be the setup used initially when headless setup is not configured yet.
- Headless setup can be configured, i.e. remote access to RPI can be done via Wifi or Ethernet. Protocol used could be SSH or VNC.
- Once headless setup is configured, the keyboard and mouse is not required anymore. Monitor can still be kept for visualization purpose.



Devices Interface (RPI and PC)

- Connection is via **Wifi**
- In the basic setup, **RPI** is configured as **Wifi AP** with **static IP**. PC is connected to the RPI wifi and issued with an IP address from the **DHCP server** in RPI.
- Note that devices in the RPI Wifi network will not have internet access.
- For devices in RPI's Wifi network to have internet access, one possible way is for RPI to be connected to the internet via **Ethernet**, and configure RPI to forward WLAN data to ETH, and ETH data to WLAN. Note that this is required only during development phase.
- The above setup are detailed in the **RPI_InfraSetup** guide in NTULearn.



Devices Interface (RPI and Tablet)

- Connection is via **Bluetooth Classic**
- RPI is **paired** with the Android tablet via Bluetooth.
- A connection is established between the RPI and Tablet via Bluetooth Serial Port Profile (**SPP**) .
- On RPI side, a **RFCOMM** channel is setup to connect to the Tablet.
- After connection is established, data can be transferred between RPI and Tablet using RFCOMM protocol via **socket programming**.
- The above setup are detailed in the **RPI_Infra Setup** guide in NTULearn.

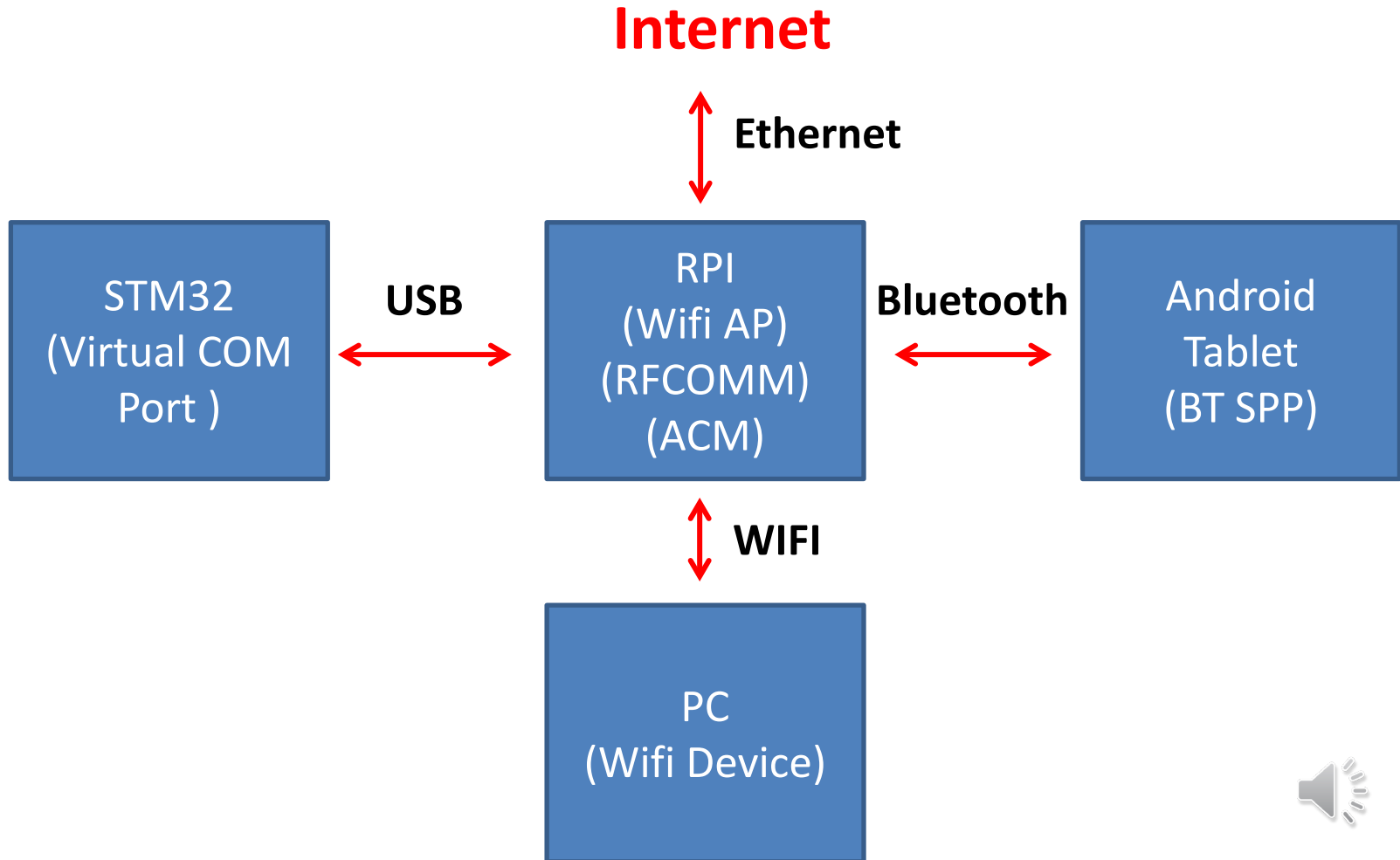


Devices Interface (RPI and STM32)

- Connection is via **USB**
- To be more specific, RPI is using **ACM** (CDC) protocol over USB to communicate with STM32.
- From software perspective, communication is via UART/ACM protocol after setup has been done.
- The above setup are detailed in the **RPI_Infra Setup** guide in NTULearn.

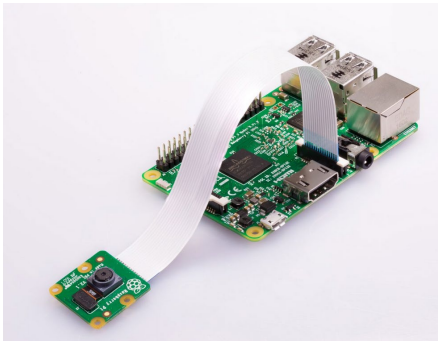


Final System Diagram



RPI Camera

- The RPI camera module used is **v2.1**.
- Note that the **Python Picamera module** is currently not, by default, compatible with the latest version of Raspberry Pi OS (**Bullseye**). To use the Picamera module, user need to **enable legacy support** for the camera. More details in the RPI setup guide.
- The Camera is connected to the RPI via a **ribbon cable**. Check the figure below for the **orientation** of the cable.
- Many libraries available for access to camera.



```
from picamera import PiCamera
from time import sleep

camera = PiCamera()

camera.start_preview()
sleep(5)
camera.stop_preview()
```



Image Recognition

- Both tasks (**Image recognition** and **Fastest Car**) requires the robot to have image recognition capability.
- There are many packages and framework that can be used for image recognition purpose.
- We used to recommend **OpenCV** image processing library, but students are leaning towards using **DNN/CNN** models.
- An introduction guide has been drafted for **TensorFlow** and is available in NTULearn.
- Setup guide for **OpenCV** is also included for students who wish to use the image processing libraries.



Examples of Image List for Recognition

- Note that this may not be the latest image set or IDs. Please check the latest MDP briefing slides for updated version of the images.

description	image	Image id	description	image	Image id
Up arrow		1	four		9
down arrow		2	five		10
right arrow		3	Alphabet A		11
left arrow		4	Alphabet B		12
Stop		5	Alphabet C		13
one		6	Alphabet D		14
two		7	Alphabet E		15
three		8			

Image with
Obstacles



Guides available in NTULearn

- RPI system setup.
- TensorFlow Image recognition.
- OpenCV setup guide **(optional)**.
- RPI Multi-threading document and lab manual.



RPI Wifi AP SSID and IP Address

- Ensure that every team uses **unique SSID** and **different static IP** range for the RPI Wifi AP server. Using same ID or IP will result in clashes.
- Suggestion
 - SSID
 - MDPGrp1 for Grp#1,
 - MDPGrp**12** for **Grp#12**.
 - Static IP Range
 - 192.168.1.1 for Grp#1,
 - 192.168.**8**.1 for **Grp#8**



Code Development

- PC processor is **x86-based** so executables compiled cannot run on RPI processor, which is **ARM-based**,
- Will need a **cross compiler** to generate ARM executables.
- **Easier** to develop the code directly on RPI.



Programming Language

- Which programming language to use depends on the purpose and familiarity.
- **C language**
 - Most **efficient**.
 - Supports **Multi-threading**.
- **Python**
 - Most are more **familiar** with Python.
 - But note the performance differences between C and Python.
 - Multi-threading is better supported in C than Python.



Interfacing protocols - Recap

- **RPI and PC**

- Wifi
- RPI => Wifi AP, PC=> Wifi Device
- IP socket programming (server-client)

- **RPI and Tablet**

- Serial Port Profile over Bluetooth
- RPI (Rfcomm)
- Tablet (SPP)

- **RPI and STM32**

- ACM/CDC over USB
- RPI (ACM)
- STM32 (ACM/CDC/UART) depending on the libraries used.



Multi-tasking Design

- RPI is handling **three communication interfaces** with PC, Tablet and STM32.
- **Options available**
 - **One program** looping three interfaces sequentially.
 - **Three separate programs** multi-tasked by RPI OS.
 - How to pass information between the programs? Pipes?
 - **One multi-thread program.**
 - Spawn three threads, one for each interface.
 - Inter-thread communication is **simpler** as they share the **global memory**. Compared to inter-process communication.
 - But may need **synchronisation** and **critical section protection**.
 - More details in the Multi-thread document in NTULearn



A little on Bluetooth ...

- Bluetooth used is the **Classic BT** and not the Bluetooth Low Energy protocol.
- In Bluetooth, a network is formed between a **Master** and max of **7 slaves**, known as a **piconet**.
- The **Pairing** process executed binds the master and the slave.
- After pairing, a **transport protocol** is needed for data transfer between devices. This is where the **SPP** and **RFCOMM** is used.
- In the setup guide, the **RPI** is configured as the RFCOMM **Server** (listening for request) while the Android **Phone** is the RFCOMM **Client**.
- In a Client-Server architecture, the Server provides the service that the Client requested.
- The RFCOMM “connection” related procedures executed in the setup guide performs the Server-Client discovery and configuration steps.
- More details in the setup guide.



TensorFlow

- **TensorFlow** is introduced as one of the of the framework students can use for **CNN model** training.
- An introductory guide is written for Tensorflow and available in NTULearn.
- It is **not mandatory** to use TensorFlow.
- But for **quiz purpose**, we need to define a smaller scope to make the load manageable.
- **Tensorflow will be used as a reference for quiz questions** where NN framework is concerned.



Quiz

- **Online. Open book. 30 mins.**
- MCQ, Multiple Answers and Short Answer (**many of them**)
- **Penalty marks** for wrong answers in Multiple-Ans type questions so do not click excessively.
- The **entire list** of questions will can be viewed in one page.
- Note that the **number of questions** are designed such that it may be **challenging to complete** all in the stipulated time.
- Topics include **RPI setup**, **Multi-Threading** and **Image Recognition**.
- Image recognition scope restricted to NN model approach for image recognition, with **Tensorflow** as the framework.
- The materials in the RPI setup, Multi-threading and Image recognition guide is a good **reference** point.

